

# Linear Data Structures: Linked List

لنکڈ لسٹ زنجیر کی طرح نوڈز پر مشتمل ہے جو میموری میں بکھرے ہو سکتے ہیں۔

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 16 of 24

## Linear Data Structures: Linked List

### LEARN IT

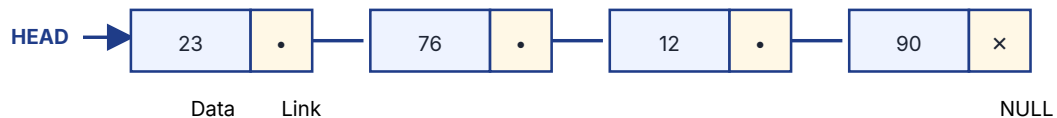
A **linked list** stores data as a chain. Unlike an array, the items are **not** one continuous block. Each separate piece is a **node** sitting somewhere in memory.

Each node has two parts:

1. the actual **data**;
2. a **link** (pointer) holding the memory address of the **next** node.

The first node is pointed to by **HEAD**. The last node's link is **NULL** (nowhere). To find an item you start at HEAD and follow links one by one — **sequential access**,  $O(n)$ .

**Real-life example:** browser history. Clicking Back follows the pointer to the previous page-node.



Follow HEAD → 23 → 76 → 12 → 90 → NULL

Four nodes. The last link is NULL, which ends the chain.

### WORKED EXAMPLE

**Worked example:** walk from HEAD to print every data field.

```
nodes = [23, 76, 12, 90]
for data in nodes:
    print(data, end=" → ")
print("NULL")
```

#### OUTPUT

23 → 76 → 12 → 90 → NULL

There is no `nodes[2]` jump in a real linked list. The loop above models following each link once. Inserting a new node between 76 and 12 only means changing two pointers — you do not shift the rest of a big block, which is why lists beat arrays for frequent insert/delete.

### REMEMBER IT

Array = row of lockers with numbers on the doors. Linked list = treasure hunt: each clue (link) tells you where the next box is.

### COMMON MISTAKE

**Mistake:** writing `list[3]` and calling it linked-list access.

**Fix:** a true linked list has no index. Python's `list` type is an array-like structure. In this chapter, "linked list" means HEAD-and-pointers.

### اردو میں سمجھیں

لنکڈ لسٹ زنجیر کی طرح نوڈز پر مشتمل ہے جو میموری میں بکھرے ہو سکتے ہیں۔ ہر نوڈ میں ڈیٹا اور اگلے نوڈ کا پتہ (Link) ہوتا ہے۔ پہلا نوڈ HEAD سے ملتا ہے، آخری کا لنک NULL ہوتا ہے۔ براؤزر کا Back بٹن پچھلے نوڈ کے پوائنٹر پر چلتا ہے۔ رسائی ترتیب وار ہے، انڈیکس سے چھلانگ نہیں۔

### CHECK YOURSELF

1. Name the two parts of a node.
2. What does the last node's link store?
3. Why is browser history a fair analogy?

Answers: 1. Data, and a link/pointer to the next node. 2. NULL (no next node). 3. Back follows the pointer to the previous page-node instead of jumping by index.

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Next lecture: Linear Data Structures: Stack